Series 4

Even though this PVK is structured into thematic sections and some content has not yet been covered, it is explicitly permitted and *should* be the case that in the problem sets *all the material* covered in the semester and in the lecture may be used.

Exercise 1 (FS2024 A.Figalli)
Let $f : [0, \infty) \to \mathbb{R}$ be a continuous function such that $f(0) = 0$ and the limit $L = \lim_{x\to\infty} f(x)$ exists.

(a) Show that f is bounded.

(b) Let m be the infimum of f on $\mathbb{R}$. Show that $m \leq 0$.

(c) Suppose that $L \geq 0$, show that f attains its minimum.

(d) Show that f does not necessarily have to attain its minimum if $L < 0$.

Exercise 2
Solve the initial value problem:

$$\begin{cases} y'(x^2+1) - 2xy = 0, \\ y(0) = 1. \end{cases}$$

Exercise 3 (FS2024, A. Figalli)
Solve the initial value problem:

$$\begin{cases} u'' + u' - 2u = -2x, \\ u(0) = \frac{3}{2}, \\ u'(0) = 2. \end{cases}$$

Exercise 4 (FS2024, A. Figalli)
Compute the integral

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx.$$

Exercise 5 (FS2024, A. Figalli)
Compute the integral

$$\int \frac{2x+1}{x(x-1)} dx.$$

Exercise 6 (FS2024, A. Figalli)
Compute the integral

$$\int \frac{\sin(x)}{\cos(x)+1} dx.$$

Exercise 7 (HS2023, A. Figalli)
Compute the Taylor polynomial at 0 up to third degree of the function

$$\cos^2(x) - \cos(x^2)e^{x^2}.$$

Use Big-O Landau notation to indicate the remainder term.

Exercise 8 (FS2024, A. Figalli)
Compute the Taylor polynomial at 0 up to fourth degree of the function

$$\sin(x^2) - \sin^2(x).$$

Use Big-O Landau notation to indicate the remainder term.

Exercise 9
Compute the integral

$$\int \log(1 + x^2) dx.$$

Tip: Further integrals for practice can be found in the Moodle integral trainer and in the exam collection. Further ODEs can be found in series 13 (HS2025).